
DO LANGUAGE MODELS FORECAST VALUE CHANGE, OR RECALL IT? A PRE-REGISTERED EVALUATION ON THE 2024 GENERAL SOCIAL SURVEY

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Abstract

Whether machine systems can forecast the trajectory of human values is an empirically testable question with direct relevance to AI alignment: any proposal to align systems toward where values are heading needs the forecasting step to be accurate and, above all, honestly uncertain. We report a pre-registered evaluation against the 2024 General Social Survey: 20 attitude items, forecast from survey-weighted history through 2022, with training-data contamination controlled by verified model cutoffs and the analysis plan committed before any model call. The 2024 wave proved unusually informative — 8 of 20 items reversed their pre-2022 trend, four by the largest single-wave decline in their recorded series. Every arm’s point forecast for the same-sex-relations item landed between 60.2 and 66.5 against an actual of 55.9; last-value persistence beat every language model on accuracy (MAE 3.15 vs. 3.58–4.07 across pre-registered arms, $n = 20$); and clean-arm 90% intervals covered 50–55% of actuals while the naive and linear baselines’ intervals covered 90%. An anonymization probe locates the “LLM forecasting skill” reported by identity-conditioned backtests in recall rather than dynamics: naming the item moved one model’s long-horizon forecast by 30.5 points, and its edge over the best classical baseline disappeared anonymized — the design contrast that turns a naive pilot’s 2.2× “LLMs beat baselines” headline into this paper’s null. We recommend anonymized-series reporting and strictly forward pre-registration as protocol standards, note that vendor model retirements are closing the window for retrospective clean evaluation, and outline forward-registered arms for GSS 2026 and 2028.

Keywords value forecasting, AI alignment, data contamination, calibration, General Social Survey

1 Introduction

A recurring proposal in AI alignment is to point systems not at a snapshot of present human preferences but at where human values are heading — post-reflection, post-deliberation, or simply later [Danaher, 2021, MacAskill et al., 2020]. Every version of that proposal contains an empirical subroutine: something must *forecast* value change, and if that step is unreliable — or worse, confidently wrong — alignment targets built on it inherit the failure. Unlike most alignment desiderata, this one is directly gradeable against half a century of resolved value measurement [Gneiting and Raftery, 2007]. Large language models look like natural candidates: they have absorbed most of the written record of value change, and a growing “silicon sampling” literature reports that they reproduce human survey responses [Argyle et al., 2023, Santurkar et al., 2023, Durmus et al., 2023] and, in some studies, predict them. But the same property makes them treacherous to evaluate — a model that has *read about* the liberalization of American attitudes can reproduce it on demand, and a backtest that asks a model to “predict” survey waves inside its training window cannot distinguish forecasting from recall [Roberts et al., 2023]. Geopolitical LLM forecasting adopted a strictly forward evaluation standard for exactly this reason [Halawi et al., 2024]; value forecasting has not.

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This paper reports what we believe is the first pre-registered, contamination-controlled evaluation of value forecasting with current models. The design was committed to a public repository before any forecasting API call (commit `0ec5fd3`, 2026-07-10), with falsifiable expectations stated in advance. The holdout is the 2024 wave of the General Social Survey (GSS), whose microdata first became public in 2025 — after the training cutoff of every “clean” model arm, as documented by each vendor — and it proved an unusually severe test: 8 of the 20 items we track reversed their pre-2022 trend, four posting the largest single-wave decline in their item’s recorded history.

Three results follow: nothing beat last-value persistence on the aligned two-year horizon (best clean LLM MAE 3.92 vs. naive 3.15, $n = 20$), and the clean LLMs erred more on the eight reversal items than persistence did (6.1–6.3 vs. 5.3 points); the calibration failure was asymmetric (clean LLM 90% intervals covered 50–55% of actuals; naive and linear intervals, 90%); and an anonymization probe shows the apparent LLM skill reported by identity-conditioned designs to be largely recall — naming the item moved one model’s long-horizon “forecast” by 30.5 points and flipped the sign of its edge over the best classical baseline. We close with protocol recommendations, a note on the closing window for retrospective clean evaluation, and a planned forward registration against GSS 2026 and 2028.

2 Related work

Silicon sampling, forecasting, and contamination. Argyle et al. [2023] introduced LLMs conditioned on demographic personas as simulated survey samples; follow-on work measures whose opinions models reflect within the United States [Santurkar et al., 2023] and across countries [Durmus et al., 2023]. That literature mostly evaluates *reproduction* of measured opinion inside the model’s training window — and when the target sits inside the training window and the item is named, reproduction and forecasting are observationally confounded. Geopolitical LLM forecasting solved the analogous problem by evaluating only on questions that resolve after data collection [Halawi et al., 2024]; we apply that discipline to value change, where survey resolution lags make strictly forward designs slow and a verified-cutoff retrospective design the practical intermediate. Roberts et al. [2023] exploit training cutoffs as a natural experiment and document benchmark performance tracking a problem’s presence in training data rather than its difficulty; our anonymization probe is the survey-series analogue, measuring per item how much of a “forecast” is attributable to the item’s name.

Calibration and scoring. Proper scoring rules make honest probabilistic reporting the unique optimum [Gneiting and Raftery, 2007], and the alignment-relevant property of a value forecast is less its point accuracy than its calibration: a system acting on a 90% interval that covers half the time is confidently wrong about the values it is optimizing around. Documented sycophancy pressure in preference-trained models [Sharma et al., 2024] is one mechanism for systematically narrow, agreeable intervals; scoring against resolved outcomes removes that gradient, though it does not constrain off-distribution behavior and inherits the performativity of any published social forecast [Perdomo et al., 2020].

3 Pre-registered design

The full plan, including five falsifiable expectations scored in the results below, was committed before any forecasting call (`ea-rewrite-2026-07/PREANALYSIS_PLAN.md`, commit `0ec5fd3`); results landed in later commits (first: `5823b96`). We summarize it here.

3.1 Data

We use the GSS cumulative microdata file, 1972–2024, Release 2 [Davern et al., 2025], with survey weights (`wtssall` for years ≤ 2018 ; `wtssnrps`, NORC’s recommended non-response-adjusted weight, for 2021–2024). For each item we compute the weighted percent giving a pre-specified target response among substantive responses, requiring at least 50 unweighted valid responses per wave, at least 6 waves, and presence in the 2024 wave. Of a pre-registered 22-item candidate set, 21 items met the extraction gates and 20 have a 2024 wave (SPKHOMO does not), so $n = 20$ throughout. The set deliberately mixes monotonic (e.g., HOMOSEX, GRASS), stable (e.g., POLVIEWS, GUNLAW), and thermostatic (e.g., EQWLTH, NATHEAL) trajectories. A pre-registered sanity gate — the computed HOMOSEX “not wrong at all” share must land near 61% in 2022 and 55% in 2024 — guards the weight and recode logic (computed: 62.7 and 55.9).

Holdout. GSS 2024 was fielded April–December 2024; its first public microdata release was in 2025. A model whose vendor-stated training cutoff predates the end of fieldwork therefore cannot have seen any 2024 response.

3.2 Horizons

Primary, aligned: history through 2022 → forecast 2024. All tested models’ training knowledge covers published GSS results through 2022, so shown history matches each model’s knowledge horizon: no arm can gain by recalling waves it was not shown, and the comparison with classical baselines is information-fair. **Long, deliberately confounded:** history through 2010 → forecast 2024, where each model’s training covers 2012–2022 outcomes it was *not shown*. The long horizon exists to measure the recall effect — it is dissected by the anonymization probe, not used to rank forecasting skill.

3.3 Arms, baselines, and probes

Three pre-registered model arms, with vendor-stated training cutoffs recorded (with source URLs) in the committed results files: **clean, headline** — gpt-5-mini (cutoff 2024-05-31; reasoning_effort=minimal, seed 1930); **clean, robustness** — gpt-4o (cutoff 2023-10; temperature 0, seed 1930); **contaminated ceiling** — claude-opus-4-8 (cutoff 2026-01; default settings). Each forecast is a single draw returning JSON {point, lo90, hi90}; the prompt contains the weighted series by year plus, in the identified condition, the GSS item description, and instructs the model to reason only from the shown series. The contaminated arm is a deliberate internal check: its training window includes the GSS 2024 release itself, so a large accuracy edge — especially on reversals — would flag memorization leaking through the “reason only from the series” instruction.

Baselines, computed on the logit scale with 90% intervals: naive (last value, random-walk interval), linear (OLS, t-based interval), ARIMA(1,1,0) (model interval), and ETS/Holt. A statsmodels 0.14.6 bug in the ETS interval extraction made the coded fallback return the naive forecast for all 20 items; ETS therefore duplicates naive throughout — reported rather than patched post hoc — leaving three distinct baselines.

Probes. (a) *Anonymized series*: the headline clean model re-run on every item with identity stripped — no name or description, just “an anonymized US national survey attitude item” and the numbers — at both cutoffs. (b) *Distribution arm*: full response-category distributions for HOMOSEX, PREMARSX, and POLVIEWS, scored by total-variation (TV) distance.

Metrics. MAE, mean signed error (bias), 90% interval coverage and mean width, and MAE within ex-post 2024-behavior strata (reversal / continuation / stable), with n inline everywhere. Budget: a hard \$25 cap; actual pre-registered spend \$0.19 across 166 calls.

4 Results

4.1 GSS 2024 reversed eight of twenty trends

A mechanical, pre-specified rule classifies each item’s 2024 behavior relative to its recent trend: an item whose last five waves through 2022 trended up by more than 0.2 points/year but fell more than 2 points in 2024 (or the mirror image) is a *reversal*; |change| ≤ 2 points is *stable*; the rest are *continuations*. Eight items reversed: PRAYER (approve school-prayer ban, -8.4 points), HOMOSEX (same-sex relations not wrong at all, -6.8), NATRACE (spending too little on Black Americans, -6.7), FEFAM (reject traditional gender roles, -6.5), GRASS (marijuana legal, -5.7), PREMARSX (-3.3), FEPOL (-2.8), and POLVIEWS (liberal identification, -2.0). Three continued (health spending +5.5, interpersonal fairness -3.6, welfare spending +2.7) and nine were stable; the five largest moves in the wave were all reversals.

Four of the eight reversals are the largest single-wave decline ever recorded for that item: PRAYER (-8.4 against a prior record of -6.9 across 27 waves since 1974), HOMOSEX (-6.8 vs. -2.5, 30 waves since 1973), FEFAM (-6.5 vs. -6.0, 24 waves since 1977), and GRASS (-5.7 vs. -4.8, 29 waves since 1973). The HOMOSEX series illustrates the severity: the share saying same-sex relations are “not wrong at all” rose from 11.4% (1973) to 62.7% (2022) with no wave-over-wave drop larger than 2.5 points, then fell to 55.9% (unweighted valid n = 2,125). External polling corroborates the direction: PRRI measured same-sex-marriage support falling 69% to 67% over 2022–2023, with Republican support for nondiscrimination protections down 7 points (66% to 59%) [PRRI, 2024]. GSS methodology also changed after 2018 (mixed web and in-person modes), which NORC flags as a possible contributor to level shifts — though a mode artifact would have to be selective to produce trend-consistent rises on some items and record declines on others in the same wave.

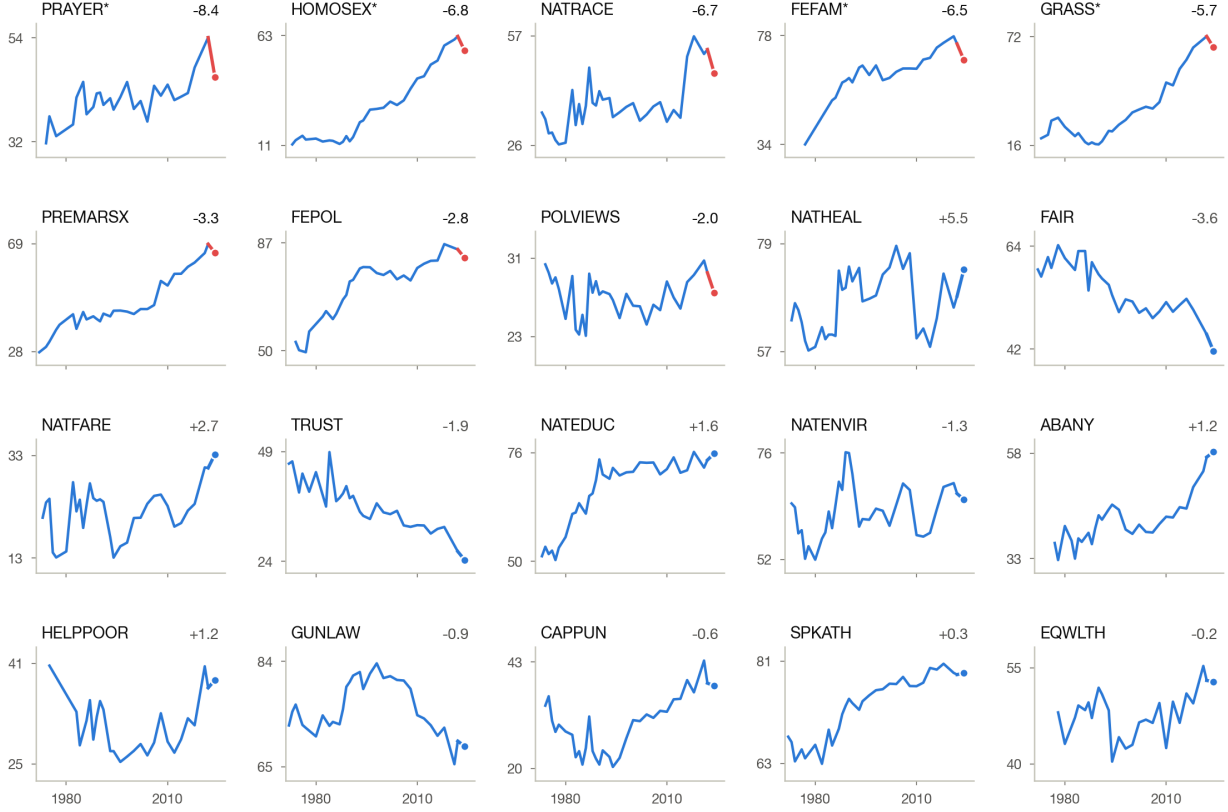


Figure 1: The 20 GSS series, 1972–2024 (survey-weighted percent giving the target response; free y-scales). Red marks the 2022→2024 segment for the eight trend reversals; annotations give the 2022→2024 change in points; asterisks mark the four largest-ever single-wave declines.

Table 1: Aligned horizon (history $\leq 2022 \rightarrow 2024$, $n = 20$). Coverage is the fraction of items whose 2024 actual fell inside the arm’s 90% interval; width is the mean interval width in points; the final column is MAE over the 8 reversal items. ETS duplicated naive for all items (see text) and is omitted.

Arm	MAE	Bias	90% coverage	Mean width	MAE, reversals
naive (last value)	3.15	+1.91	0.90 (18/20)	12.0	5.26
linear	4.07	+0.03	0.90 (18/20)	15.1	3.41
ARIMA(1,1,0)	3.58	+2.22	0.80 (16/20)	11.6	5.28
gpt-5-mini (clean)	4.07	+2.20	0.55 (11/20)	8.6	6.30
gpt-4o (clean)	3.92	+2.50	0.50 (10/20)	7.6	6.11
claude-opus-4-8 (contam.)	3.58	+1.81	0.80 (16/20)	11.0	5.42

4.2 Accuracy and calibration at the aligned horizon

Table 1 reports the pre-registered arms at the aligned horizon (history $\leq 2022 \rightarrow 2024$, $n = 20$); Figure 2 adds the post-registration robustness arms.

Three findings.

1. **Nothing beat persistence.** The best clean LLM (gpt-4o, 3.92) trailed the last-value baseline (3.15), $n = 20$; the headline clean arm’s ratio to the best simple baseline was 1.29. Every arm except

Table 2: HOMOSEX (“not wrong at all”), history through 2022 → 2024. Actual: 55.9 (weighted; unweighted valid $n = 2,125$).

Forecaster	Point	90% interval	Covers 55.9?
naive	62.7	[56.8, 68.3]	no
linear	60.2	[49.2, 70.3]	yes — width 21.1
ARIMA(1,1,0)	63.1	[56.4, 69.3]	no
gpt-5-mini (clean)	66.5	[62.0, 70.5]	no
gpt-4o (clean)	64.5	[61.0, 68.0]	no
claude-opus-4-8 (contam.)	64.0	[59.0, 69.0]	no

linear carried positive bias — the signature of extrapolating liberalization into a wave that reversed it.

2. **The LLMs missed reversals by more than persistence did** (6.1–6.3 points on the eight reversal items vs. naive’s 5.3), with no compensating edge on stable items (1.6–1.7 vs. 1.0).
3. **LLM intervals were confidently wrong.** Clean-arm 90% intervals averaged 7.6–8.6 points wide and covered 50–55%; classical intervals averaged 11.6–15.1 points and covered at or near nominal (naive and linear exactly 18/20; ARIMA 16/20). On this task, knowing that you don’t know was worth more than any dynamics the models had learned.

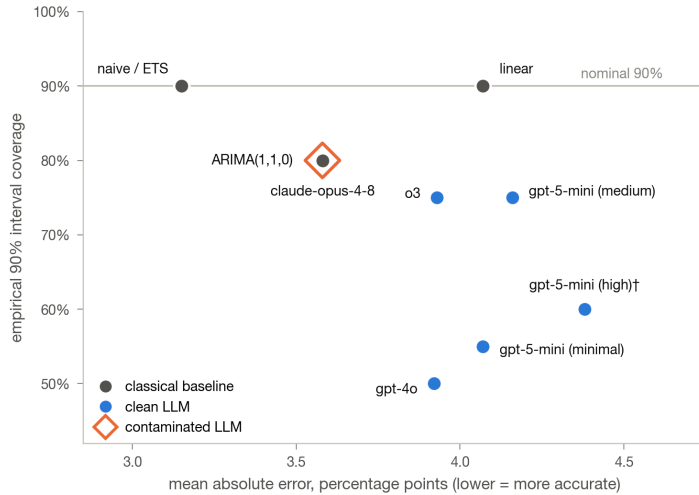


Figure 2: Accuracy versus calibration, aligned horizon ($n = 20$ unless marked). Classical baselines sit at or near nominal coverage; every LLM arm — including the post-registration o3 and effort-sweep arms — sits below it. The high-effort arm completed 10 of 20 items. ARIMA and claude-opus-4-8 coincide at (3.58, 0.80); ETS duplicated naive and is not shown.

4.3 The reversal everyone missed

Every point forecast landed between 60.2 and 66.5 against an actual of 55.9 (Table 2); the only interval containing the truth (linear) did so by being 21 points wide, not by anticipating a downturn, and the anonymized clean run’s 67.0 [61, 73] misses too — at this horizon the failure is dynamics, not item identity. Across all eight reversal items, 90% intervals covered the actual 6/8 (naive), 7/8 (linear), and 6/8 (ARIMA) times, against 1/8 (gpt-5-mini), 0/8 (gpt-4o), and 4/8 (claude-opus-4-8).

4.4 Where identified “skill” comes from: the anonymization probe

At the aligned horizon, stripping item identity barely matters: the mean |identified - anonymized| point difference is 1.0 point ($n = 20$), and the model trails naive either way (4.07 identified, 4.62 anonymized, vs.

3.15). At the 2010 cutoff — 14 years of un-shown history inside the model’s training data — identity moves forecasts a mean of 4.5 points, and the items that move most are the famous ones (Figure 3): HOMOSEX +30.5 (78.0 identified vs. 47.5 anonymized, actual 55.9), GRASS +17.0, PREMARSX +9.5, FEFAM +9.0.

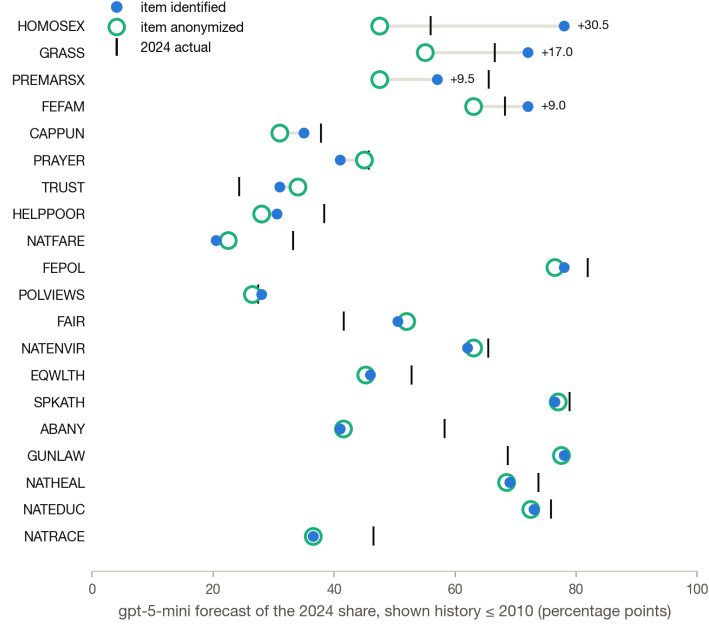


Figure 3: The anonymization probe at the 2010 cutoff: gpt-5-mini’s 2024 forecast per item, identified (filled) vs. anonymized (ring), sorted by the gap (labeled where ≥ 5 points). The four largest gaps are the set’s best-known liberalization trajectories; where the marks coincide, the name added nothing.

The HOMOSEX case rewards attention. Shown data through 2010 (42.7%) and told to reason only from the series, the named model answered 78 — a level the GSS series has never reached, 15 points above the 2022 value in its training data; the name appears to have pulled in the level of adjacent, differently-worded series (Gallup’s acceptance questions run well above the GSS’s four-category “not wrong at all” share), a confusion worth auditing for in any identified-series design. Anonymized, the same model extrapolated the same numbers to 47.5. Actual: 55.9. Both wrong, 30.5 points apart.

In aggregate at the 2010 cutoff, the identified model beats naive by 1.30 points of MAE (7.25 vs. 8.55, $n = 20$); anonymized, the edge shrinks to 0.82 (7.73 vs. 8.55). Against the strongest classical baseline at that horizon (ARIMA, 7.66), the identified edge of +0.41 flips to -0.07 under anonymization. We had pre-registered “the edge collapses under anonymization”; the honest reading is *partial* — the edge over naive shrinks by roughly a third, the edge over the best baseline disappears, and individual identified forecasts move by up to 30.5 points on the name alone.

The contaminated arm tells the same story from the other side. At the aligned horizon claude-opus-4-8 shows no memorization signature: MAE 3.58 vs. the clean arm’s 4.07 ($n = 20$), a miss on HOMOSEX like everyone else, 4/8 reversal coverage. At the 2010 cutoff its advantage is exactly what recall of un-shown history predicts: MAE 4.18 vs. the clean arms’ 5.83–7.25 ($n = 20$), with reversal-item MAE 2.89 vs. their 4.22–7.39 ($n = 8$) — it “forecasts” history it was not shown because it has read about it.

Finally, the distribution arm: asked for full response-category distributions for 2024, both tested models produced shapes close to the truth — total-variation distance 0.081 on HOMOSEX (both), 0.054/0.043 on PREMARSX (gpt-5-mini/claude), 0.021/0.027 on POLVIEWS — while putting the top category of the two sexual-morality items 8.1 and 3.4–5.4 points too high: the shape of heterogeneity is roughly right, and the epistemic error concentrates in the same direction as everywhere else.

Table 3: Post-registration robustness arms, aligned horizon (vendor-stated cutoffs: o3 2024-06-01, gpt-5-mini 2024-05-31 — both clean). Reference points from the pre-registered record: naive 3.15 MAE / 0.90 coverage; gpt-5-mini (minimal) 4.07 / 0.55. The high-effort arm completed 10 of 20 items before a session crash; its metrics are over that subset.

Arm	MAE (n)	90% coverage	MAE, reversals	HOMOSEX (actual 55.9)
o3, default effort	3.93 (20)	0.75 (15/20)	6.26	65.0 [59.0, 71.0] — miss
gpt-5-mini, medium effort	4.16 (20)	0.75 (15/20)	7.11	65.2 [60.0, 70.4] — miss
gpt-5-mini, high effort	4.38 (10)	0.60 (6/10)	7.03	66.3 [62.1, 70.5] — miss

4.5 Post-registration robustness arms

Three anticipated objections were checkable cheaply, so we ran them after registration as labeled robustness arms — separate result files, the pre-registered record untouched; same items, prompts, and metrics, at the aligned horizon.

“Use a reasoning model.” o3 — the strongest reasoning model available to us whose stated cutoff predates the end of GSS 2024 fieldwork — improves calibration on the smaller clean arms’ 0.50–0.55 (to 0.75) but stays overconfident, trails persistence (3.93 vs. 3.15, $n = 20$), and misses the reversal.

“You ran the LLM at minimal effort.” Raising gpt-5-mini’s reasoning effort made accuracy slightly worse: 4.07 (minimal, $n = 20$) $\rightarrow$ 4.16 (medium, $n = 20$) $\rightarrow$ 4.38 (high; partial arm, $n = 10$). More deliberation converged on the same wrong prior (Figure 2).

“Add a clean Anthropic arm.” Impossible at evaluation time, and the reason generalizes: every Claude snapshot with a training cutoff predating the end of GSS 2024 fieldwork had been retired from the first-party API (claude-3-5-sonnet-20241022 returns 404; retired October 28, 2025 [Anthropic, 2026]). The population of models that can ever be cleanly tested on a given holdout shrinks as vendors deprecate old snapshots — retrospective clean evaluation has a **closing window**. A planned DeepSeek open-weights arm was skipped for the mirror-image reason: the vendor documents no training cutoff, so the arm could not be labeled clean at all.

The robustness arms cost roughly another dollar (measured \$0.85 for the completed high-effort segment; cost logs for the o3 and medium-effort segments were lost to a session crash, though every raw call is preserved in the committed results file).

4.6 Scorecard against the pre-registered expectations

Table 4 scores each expectation committed in the pre-analysis plan against its outcome: three confirmed, one failed as stated, one partial.

5 The result a naive design produces

Before building the controls, we ran this task the way much of the LLM-simulation literature runs it: item names shown, forecast targets inside the model’s training window, no survey weights, two variables. That pilot produced a language model beating the strongest time-series baseline it tested by $2.2\times$ on MAE (12.5 vs. 28.1, $n = 2$ items, targets 2000–2021 from cutoffs in 1990/2000) — a publication-ready number, structurally similar to results that are being published. Each of the pilot’s findings, traced through the controlled design:

- **“LLM beats best baseline $2.2\times$ ” (MAE 12.5 vs. 28.1).** Produced by $n = 2$ items, names shown, targets 2000–2021 inside the training window. Does not replicate: clean LLM 4.07 vs. naive 3.15 (ratio 1.29, LLM worse), $n = 20$.
- **“Everyone is overconfident” (LLM 43%, baselines 36% coverage).** Produced by the same design with degenerate baseline intervals. Half survives: LLMs cover 50–55% at 90% nominal; proper classical intervals cover 90%.

Table 4: The five falsifiable expectations committed before any forecasting call (commit 0ec5fd3), and their outcomes.

#	Pre-registered expectation	Outcome
1	All arms miss the HOMOSEX reversal (point ≥ 60 vs. actual ≈ 55)	Confirmed — points 60.2–66.5 vs. 55.9; linear’s 21-point interval did contain the actual
2	Overconfidence replicates: 90% coverage < 0.90 for every arm	Failed as stated — LLM arms under-covered (0.50–0.80) but naive and linear hit 0.90 exactly; only the LLMs are overconfident
3	Clean LLM within ~ 1 point of the best simple baseline; no “LLM $\gg$ baselines”	Confirmed — 4.07 vs. 3.15, $\Delta = 0.92$, LLM on the worse side
4	Contaminated $\approx$ clean at the aligned horizon; no reversal-nailing	Confirmed — 3.58 vs. 4.07; HOMOSEX missed; 4/8 reversal coverage
5	Long-horizon identified edge collapses under anonymization	Partial — edge over naive 1.30 $\rightarrow$ 0.82; edge over ARIMA +0.41 $\rightarrow$ -0.07; per-item swings up to 30.5 points

- **“LLMs capture non-linear dynamics baselines miss.”** Produced by the 1990 $\rightarrow$ 2021 HOMOSEX run — dynamics the model had read about. Under anonymization, the long-horizon edge over the best baseline disappears.
- **“Trajectories reverse and trend-followers miss it.”** Produced by GPT-4o vs. GSS 2024, run before the microdata release. Replicates, now pre-registered: 8/20 reversals, every arm missed HOMOSEX.

The same pilot included a calibrated long-horizon projection placing same-sex-relations acceptance at 66 by 2030 (80% CI [57, 75]); GSS 2024’s 55.9 already sits below that interval’s floor. A design of this shape asks a model to “forecast” years it has read about, on items it can name, and reports recall as skill. The one thing it got right — trajectories reverse and trend-extrapolators miss it — was pre-registered this time (expectation 1) and confirmed exactly.

6 Discussion

Protocol recommendations. Two changes would let the LLM-as-social-forecaster literature grade itself honestly. First, *anonymized-series reporting*: any retrospective claim that an LLM predicts opinion, elections, or social-science outcomes should report the anonymized-series number beside the identified one — the gap directly measures how much claimed skill is recall. It is cheap (one extra run) and imperfect (the dynamics themselves may be memorized). Second, *strictly forward pre-registration*, the standard geopolitical LLM forecasting already meets [Halawi et al., 2024]: no contamination argument is possible about data that does not yet exist.

The closing window. Retrospective clean evaluation additionally depends on vendors keeping old snapshots available. They do not: every Anthropic snapshot cleanly positioned for the GSS 2024 holdout had been retired from the first-party API by evaluation time [Anthropic, 2026], and open-weight alternatives without vendor-documented cutoffs cannot be labeled clean either. Each wave’s set of cleanly testable models shrinks monotonically; forward registration and open-weight models with documented provenance are the durable fixes.

What the failure means for value-forecasting-as-alignment-target. Using a forecast distribution of human values as an alignment input requires, at minimum, that the forecasting machinery beat trivial baselines and cover at its stated rate. At the two-year horizon nothing we tested clears the first bar and no LLM arm clears the second — while persistence *with honestly wide intervals* performs exactly as advertised (18/20 at 90% nominal). A system aligned in 2022 to “acceptance keeps rising, 90% sure” would have been confidently wrong, within two years, about the direction of the value it was optimizing around. The

distribution arm locates the failure as epistemic rather than structural — the *shape* of heterogeneity is roughly right ($TV \leq 0.081$) while its level slides with the broken trend — arguing for distributional targets with calibrated uncertainty, not point-trajectory extrapolation; the normative step remains human [MacAskill et al., 2020].

A contamination-proof instrument. Every retrospective design above still leans on cutoff bookkeeping, because every modern LLM has read the polling record it is tested on. Vintage-corpus models remove the problem at the root: Talkie-1930 is a 13B model pretrained exclusively on 260B tokens of pre-1931 text [Levine et al., 2026], alongside the time-locked Ranke-4B family (cutoffs 1913–1946) [Göttlich et al., 2025] and TypewriterLM (pre-1913) [Luo et al., 2026]; Underwood et al. [2025] argue period pretraining, not fine-tuning, is required to avoid hindsight. For value forecasting a 1930 cutoff is a feature: Gallup’s first national polls date to 1935 and the GSS to 1972, so the *entire quantitative polling record* is out-of-sample — ninety years of resolved actuals versus the single clean wave available here. The obstacle is elicitation, not knowledge: the Talkie-1930 base model approaches GPT-3-175B on arithmetic under log-probability scoring while its instruction-tuned variant scores 4.9% (strict, 5-shot) on GSM8K vs. 17.8% for same-size LLaMA-13B [Ghenis, 2026], and our exploratory Gallup-style probes of the chat layer hit the same wall (outputs in the repository). The natural design is *sensor mode*: read attitude signals from the vintage model via log-probability scoring under period framings; let a modern model, blinded to named post-1930 facts, generate hypotheses and calibrated forecasts; grade the pipeline on the 1936–2024 record under proper scoring rules. Whether attitude knowledge survives extraction the way arithmetic does is the open question, and the first to answer.

7 Limitations

Twenty items, one country, one survey program, one holdout wave: the 2024 reversals may reflect period shocks that say little about longer-horizon forecastability, and $n = 20$ gives no power to distinguish LLM arms from one another (naive’s edge over the clean arms is 0.8–0.9 points of MAE). One draw per forecast, with seeds where supported; sampling variance is unquantified. The headline clean arm ran at minimal reasoning effort — addressed post hoc by the effort sweep and o3 arms, one of which is partial ($n = 10$) — and the ETS baseline silently duplicated naive. GSS mode changes after 2018 may contribute level shifts to the very reversals every arm missed, though the direction-selective pattern argues against a pure mode artifact; cutoff labels rest on unauditable vendor self-reports; and the ex-post strata describe 2024 behavior, not forecast inputs.

8 Planned: forward registration for GSS 2026 and 2028

Status: planned, pending author confirmation — this section is not itself a registration. The binding registration will be a separate commit specifying items, arms, elicitation, and scoring before GSS 2026 fieldwork ends.

The natural next step is the design no contamination argument can touch: point forecasts, 90% intervals, and full response distributions for all 20 items (plus any newly qualifying items), for GSS 2026 and 2028, committed publicly before the data exist — from classical baselines, current LLMs at fixed settings, and any externally submitted system, scored by this paper’s metrics plus a proper scoring rule on the distributional forecasts [Gneiting and Raftery, 2007]. The repository issue tracker accepts external arms; open-weight submissions are especially welcome, since they remain re-runnable after vendor retirements.

9 Reproducibility

Everything needed to reproduce the tables and figures is committed to the public repository: the pre-analysis plan (0ec5fd3, committed before any forecasting call), extraction/baseline/forecast/analysis/figure code (ea-rewrite-2026-07/code/), all prompts and raw model outputs, per-arm token usage and costs, and the result JSONs from which every number in this paper is drawn (ea-rewrite-2026-07/results/). GSS microdata are redistributed by NORC [Davern et al., 2025] and are not committed; the extraction script regenerates the weighted series from the public cumulative file. Total API spend: \$0.19 for the pre-registered arms (166 calls) plus roughly \$1 for the robustness arms. Thanks to the NORC GSS team for the cumulative file and documentation; errors are mine.

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